

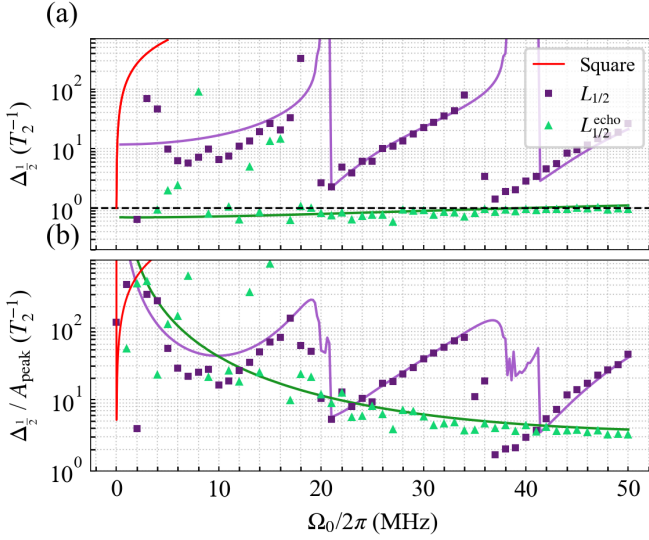


Figure 4. Comparison of spectroscopic linewidth obtained using a constant (square), Lorentzian $L^{(1/2)}$, and echo-Lorentzian $L_{\text{echo}}^{(1/2)}$ drive as a function of peak Rabi frequency Ω_0 . (a) Measured full width at half maximum (FWHM) $\Delta_{1/2}$ of the qubit spectral response, normalized to the T_2^{-1} limit, together with numerical simulations. Solid lines denote simulation results, while markers represent experimental data. The constant drive (red) exhibits conventional power broadening $\Delta_{1/2} \propto \Omega_0$, whereas the Lorentzian (green) and echo-Lorentzian (purple) pulses display power narrowing due to adiabatic following. (b) Linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$, highlighting the improved resolution-signal trade-off enabled by the echo-Lorentzian protocol at increasing drive amplitude.

level system using identical system parameters. The square (red), Lorentzian (green), and echo-Lorentzian (purple) drives are shown, where solid lines denote simulation results and markers represent experimental data. For the square pulse, only simulated results are displayed, as the agreement with experiment was independently validated. We plot both the linewidth $\Delta_{1/2}$ and the linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$ to quantify the resolution-signal trade-off.

To suppress residual power broadening at large drive amplitudes the pulse envelope must satisfy requirements on its cutoff behavior. The cutoff effectively introduces a discontinuity in the envelope derivative, effectively behave as a constant pulse, producing a linear broadening that scales with the instantaneous amplitude at the cutoff point, denoted Ω_c . For T_2 limited spectroscopy the T_2 limit $T_2^{-1} \leq \Delta_{1/2}$ therefore yields a constraint on the maximum allowable cutoff amplitude $\Omega_c < T_2^{-1}$. In our system Satisfying the above inequality requires a relative cutoff level on the order of 10^{-4} to 10^{-5} depending on pulse duration and peak drive amplitude. Empirically, we find that a cutoff less than 10^{-3} is sufficient to eliminate excess broadening and to maintain spectroscopy in the T_2 limited regime even under strong driving conditions;

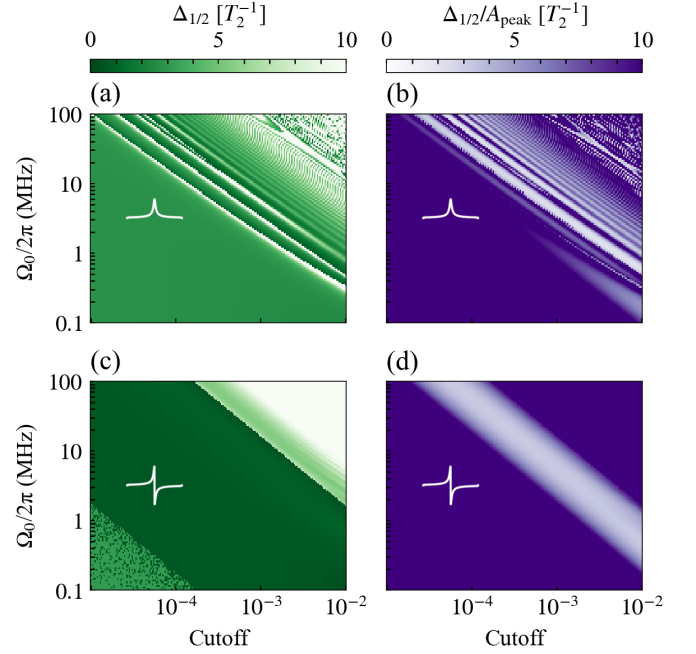


Figure 5. Numerical simulation of spectroscopy scans using root-Lorentzian $L^{1/2}$ and echo-root-Lorentzian $L_{\text{echo}}^{(1/2)}$ drives with a pulse duration of $50 \mu\text{s}$. (a),(b) Line width $\Delta_{1/2}$ and the linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$ obtained using the root-Lorentzian pulse $L^{(1/2)}$ as a function of peak Rabi frequency Ω_0 and temporal cutoff. (c),(d) Corresponding results for the echo-root-Lorentzian pulse $L_{\text{echo}}^{(1/2)}$. While $L^{(1/2)}$ pulse exhibits extended regions of narrow linewidth, the associated linewidth-to-signal ratio displays strong oscillatory behavior, making it difficult to identify robust operating parameters for high-resolution spectroscopy. In contrast, the echo variant $L_{\text{echo}}^{(1/2)}$ yields both reduced linewidth and a broad parameter region in which the resolution-signal trade-off remains favorable, enabling stable narrowband spectroscopy.

see the Supplemental Material for the cutoff analysis and supporting numerical data [18].

ROBUSTNESS AND OPERATING LIMITS

At the core of assessing pulse quality and identifying optimal operating regimes lies the spectral response of the driven qubit. We find that the echo pulse yields spectroscopic features with markedly reduced linewidths over a well-defined region of parameter space. In particular, the pulse duration must remain shorter than the qubit lifetime T_1 , while exceeding approximately $10 \mu\text{s}$ in order to suppress Fourier-limited broadening; supporting pulse-duration calculations are provided in the Supplemental Material [18].

This constitutes an additional advantage of the echo protocol. While both constant and simple Lorentzian pulses exhibit coherent Rabi oscillations that persist even for pulse durations comparable to or exceeding T_1 , the